Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination

April 2025

Max. Marks: 30 | **Duration: One Hour**

Class: TE CSE (DS/AIML) | **Semester: VI**

Course Code: AI306/DS306 | **Program: B.Tech**

Name of the Course: Distributed Computing

Instruction:

(1) All questions are compulsory

(2) Draw neat diagrams and write equations wherever necessary

(3) Assume suitable data if necessary

Course Outcomes:At the end of the course student will be able to:

(1) Understand the principles and desired properties of distributed systems. Apply the various communication techniques for distributed communication.

(2) Apply the concepts of process, naming, consistency, replication and faults tolerance in a distributed environment.

(3) Apply the algorithms such as clock synchronization, election, and mutual exclusion in distributed applications.

(4) Identify the challenges in developing distributed applications.

Q No.		Max. Marks	CO-BL
Q.1a	a. Compare RMI, MOM and RPC in terms of: (a) Communication style (Synchronous/Asynchronous) (b) Data Exchanged (c) Coupling (d) Scalability (e) Use Cases	5	CO4-BL2
	OR		
	b. Edge cloud architecture is essential for IoT and Industrial IoT (IIoT) applications due to its ability to reduce latency, optimize bandwidth, and enhance real-time decision-making. Elaborate on this statement with suitable examples and challenges.	5	CO4-BL3
Q.1b	a. Is it always necessary for the sender of a message to know that the message arrived safely at its destination? If yes, explain why. If not, give two examples in support of your answer.	5	CO1-BL2
	OR		
	b. What is "piggybacking" of a message? How does it help in reducing network traffic? Give some examples of where the piggybacking scheme may be used to improve performance in distributed systems.	5	CO1-BL2
Q.2a	How does multicast communication improve performance in distributed gaming platforms, where multiple players are connected to a game server? What advantages does multicast provide compared to unicast in terms of network efficiency?	5	CO1-BL2
Q.2b	Explain what is meant by consistent ordering and causal ordering of messages. Give a mechanism to implement each one.	5	CO2-BL2

Q.3a	"Threads play a crucial role in improving concurrency and resource utilization in distributed systems." (a) Explain the fundamental concepts of threads and how they differ from traditional processes. (b) Discuss the role of threads in distributed systems with a focus on their impact on client-server architectures.	5	CO2-BL2
Q.3b	A distributed financial application requires synchronized timestamps across multiple data centers. Compare Cristian's Algorithm and Berkeley's Algorithm for clock synchronization. Which one is better suited for this system?	5	CO3-BL3

Course outcome	**Marks in ESE**	Marks in MSE	Total marks for each course outcome
CO1		10	
CO2		10	
CO3		5	
CO4		5	